Differentiation Rules: Power, Product, Quotient, Chain

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

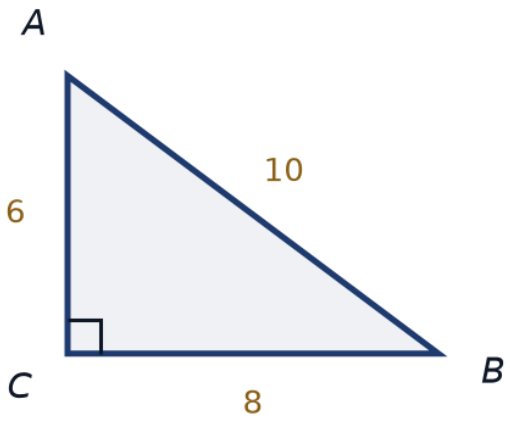
For parametric $x = e^t \cos t$, $y = e^t \sin t$, find $\frac{d^2y}{dx^2}$ at $t = 0$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin(\theta) = -\frac{24}{25}$ and θ lies in quadrant III, find $\sin 2\theta$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin(\theta) = \frac{5}{13}$ and θ lies in quadrant II, find $\cos 2\theta$.



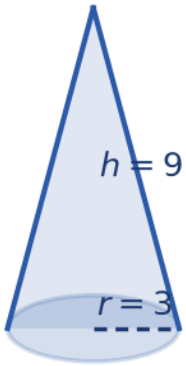
4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Coffee at 70°C is placed in a 20°C room and cools to 60°C in 10 min (Newton's law of cooling). How long (min) until it reaches 40°C?

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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Water pours into an inverted cone (radius 3 m, height 9 m) at 3 m³/min. How fast is the depth rising when it is 8 m? (m/min)



6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Product rule for $f = x^2 e^x$:.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

$\frac{d}{dx}[f(g(x))]$ equals (chain rule):.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

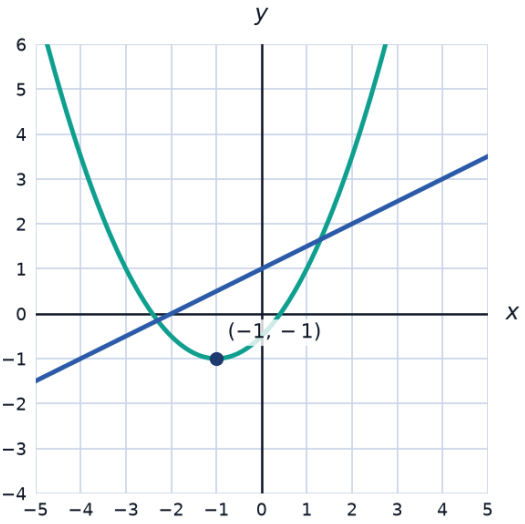
8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Quotient Rule: $\frac{d}{dx}\left(\frac{x}{x+1}\right) =$.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Product of two power series $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$: coefficient of x^n in $A(x)B(x)$ is:.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · AdvancedA differential equation $\frac{dy}{dx} = (y-1)^2 \cos(x)$ has initial condition $y(0) = 2$. Find the particular solution $y = f(x)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumThe graph of f' consists of a semicircle of radius 2 centered at the origin, lying above the x-axis. If $f(0) = 5$, find the value of $f(2)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Convergence Tests

Level 1 · FoundationsOn a Rate In / Rate Out FRQ, if rain falls into a cistern at rate $R(t)$ and is pumped out at rate $P(t)$, when is the amount of water in the cistern at a relative maximum?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 1 · FoundationsHow many total questions are on the AP Calculus AB exam, and what percentage of the composite score comes from Free-Response questions?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

[AP Calculus AB Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Double Integrals

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the derivative of $f(x) = 4x^5 - 3x^3 + 7x - 9$.

a. $20x^4 - 9x^2 + 7$

b. $20x^4 - 9x^2$

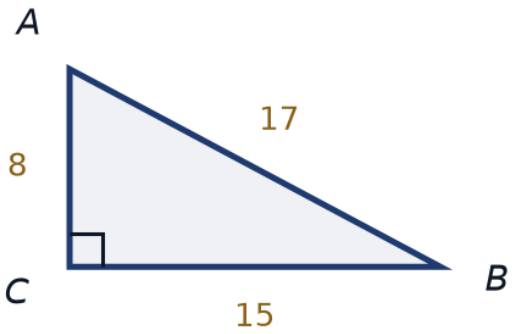
c. $4x^4 - 3x^2 + 7$

d. $20x^4 - 6x^2 + 7$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all points where the curve $y = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent.



a. $x = -1$ and $x = 3$

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the second derivative y'' for $y = \cos(2x)$.

a. $-4\cos(2x)$

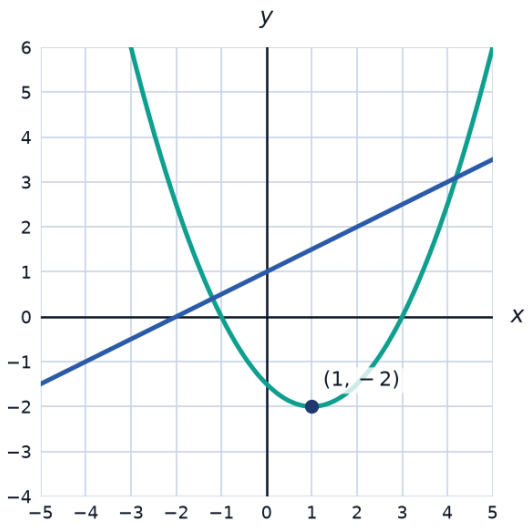
b. $4\cos(2x)$

c. $-2\sin(2x)$

d. $-4\sin(2x)$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:



a. -1

b. -2

c. 0

d. 1

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Cauchy product: $(\sum_{n=0}^{\infty} x^n)^2 = \sum_{n=0}^{\infty} (n+1)x^n$. This equals:

a. $1/(1-x)^2$

b. $1/(1-x)$

c. $(1-x)^2$

d. $2/(1-x)$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Implicit differentiation of $F(x, y) = 0$: $\frac{dy}{dx} =$

a. $-F_x/F_y$

b. F_x/F_y

c. $-F_y/F_x$

d. F_{xy}

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Logarithmic differentiation: if $y = \frac{x^2\sqrt{x+1}}{e^x}$, then $y' =$

a. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} - 1\right)$

b. $y\left(\frac{2}{x} - \frac{1}{2(x+1)} - 1\right)$

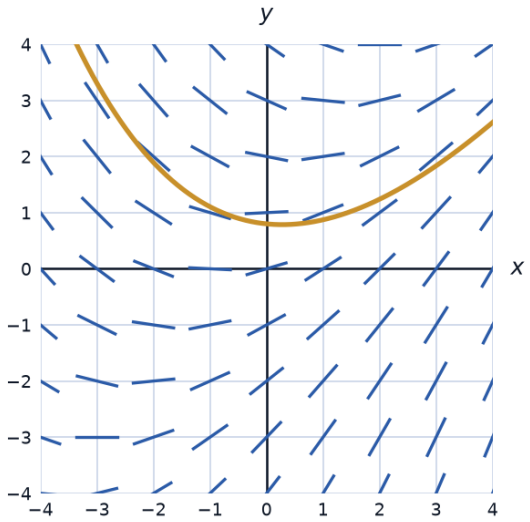
c. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} + 1\right)$

d. $y\left(2x + \frac{1}{2} - 1\right)$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Logarithmic differentiation: $y = x^x$. $y' =$



a. $x^x(1 + \ln x)$

b. $x \cdot x^{x-1}$

c. $x^x \ln x$

d. e^x

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Differentiation of geometric series $1/(1 - x) = \sum x^n$ gives $1/(1 - x)^2 =$

a. $\sum n x^{n-1}$

b. $\sum (n + 1) x^n$

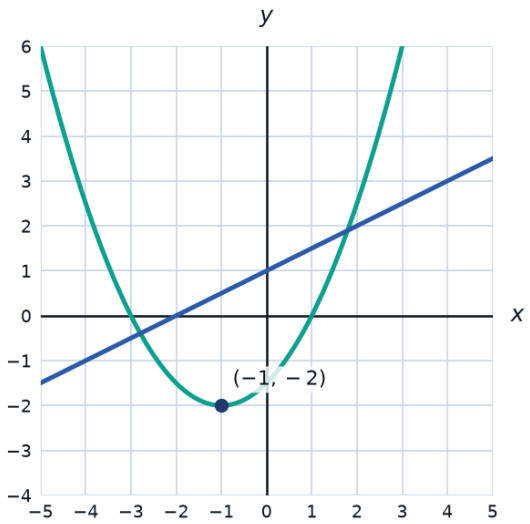
c. $\sum x^{n-1}$

d. $\sum n x^n$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Leibniz rule: $(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}$. Analogy with:



a. binomial theorem

b. FTC

c. MVT

d. Taylor series

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Term-by-term differentiation of $\sum a_n x^n$ within $|x| < R$ gives:

a. $\sum n a_n x^n$

b. $\sum a_{n+1} x^{n+1}$

c. $\sum a_n x^{n+1}/(n+1)$

d. $\sum n a_n x^{n-1}$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

L'Hôpital's rule applies to $\lim_{x \to a} \frac{f}{g}$ when the limit is of the form:

a. 0/1

b. 1/0

c. 0/0

d. 1/1

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1].

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)dx$

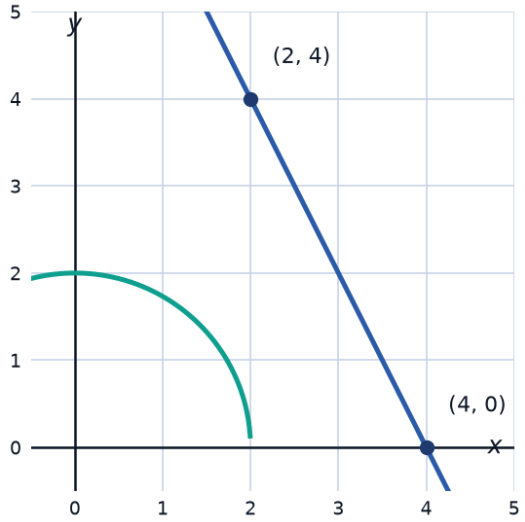
c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right)dx$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.



a. 4

b. 6

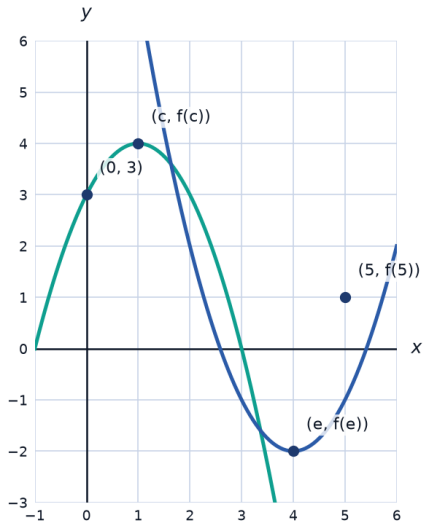
c. 5

d. 7

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36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $If\int_{0}^{c} f(x) \, dx = k, whatisthevalueof\int_{c}^{5} f(x) \, dxintermsofk$?



a. 12 + k

b. 12 - k

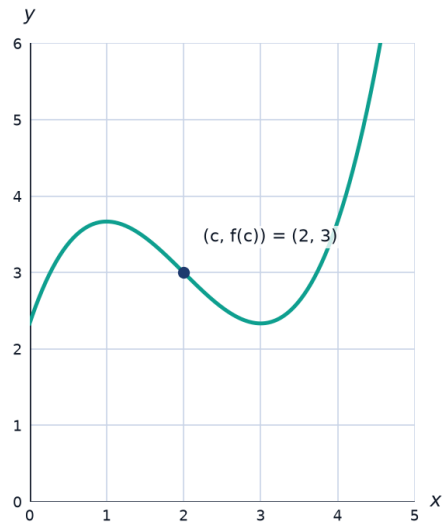
c. k - 12

d. k + 2

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37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.



a. 4

b. 3

c. 2

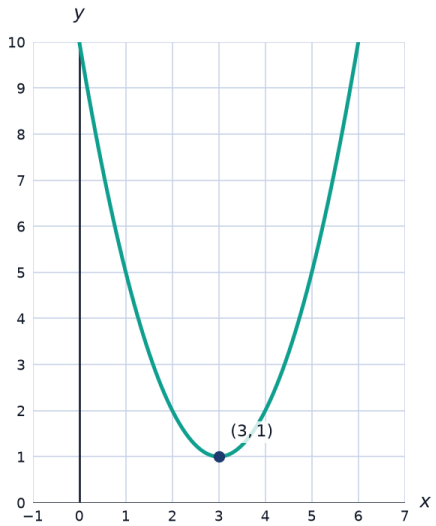
d. 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.



a. x = 3

b. x = -3 and x = 3

c. x = 1 and x = 5

d. x = 1

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Double Integrals

A differentiable function f is defined on the interval $[-4, 4]$. The graph of f', the derivative of f, is shown in the figure above. The graph of f' consists of two line segments and a semicircle. (a) Find the value of $f(2) - f(-2)$. Show the setup for your answer. (b) For how many values of c in the interval $(-4, 4)$ does $f'(c) = 0$? Explain your reasoning. (c) The function f has a local maximum at $x = k$. Find the value of k. Explain your reasoning. (d) The function f has an inflection point at $x = p$. Find the value of p. Explain your reasoning.

a. There are no values of c for which $f'(c) = 0$.

b. There is one value of c for which $f'(c) = 0$.

c. There are two values of c for which $f'(c) = 0$.

d. There are three values of c for which $f'(c) = 0$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Vector Fields and Line Integrals

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ is always positive and increasing.

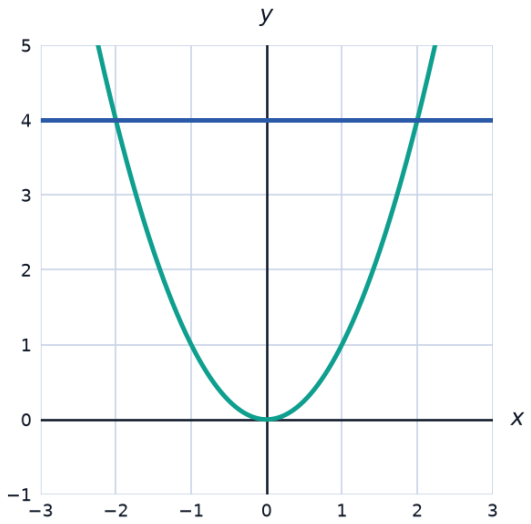
b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Evaluating Limits

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?



a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 8

d. 16

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.02 Continuity

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. 4

d. 8

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 The Derivative from First Principles

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

a. $t = \frac{3}{2}$

b. $t = 1$

c. $t = 2$

d. $t = 3$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2 + 1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$, and $g(x)$ is increasing.

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

a. g'(2) = 2

b. g'(2) = 0

c. g'(5) = -2

d. g'(5) = 0

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